

Experiment 6: Mechanistic Analysis of Grokking under Federated Learning

Federated Learning Grokking Project

March 2026

1 Overview

- **Goal:** Post-hoc mechanistic analysis of 237 FL runs from Exps 2–3 to answer: *why* does grokking survive (or fail) under federated averaging?
- **No new training runs.** All analysis uses per-round metrics already logged: test/train accuracy, loss, weight norms (layers 1 and 2), IPR, mean client drift $\bar{d}_k$, and client weight divergence σ_w .
- **Dataset:** 191 grokked runs + 46 failed runs across $\alpha \in \{0.2, 0.25, 0.3, 0.35, 0.5\}$, $K \in \{2, 5, 10, 20, 50, 97\}$, and $\alpha_{\text{dir}} \in \{0.01\text{--}1000\}$.

Four main findings:

- (i) Drift predicts grokking delay *within* each α , not merely as a proxy for α .
- (ii) IPR onset consistently precedes drift drop (72/72 runs, 12/12 threshold combinations).
- (iii) Accuracy, IPR, and normalized drift collapse onto shared curves under rescaled time.
- (iv) FL disrupts generalization but not memorization: T_{memo} is invariant to K while T_{grok} varies by up to $2.1\times$.

2 Representative Run Comparison (Figure 1)

Four runs spanning the mechanistic spectrum:

- **Healthy grok:** $\alpha=0.5$, IID, $K=10$ — fast grokking at $\sim 7,600$ steps
- **Delayed grok:** $\alpha=0.25$, $\alpha_{\text{dir}}=0.01$, $K=10$ — groks at $\sim 36,600$ steps
- **Failed (data-limited):** $\alpha=0.2$, IID, $K=10$ — fails centralized too
- **Failed (fragmented):** $\alpha=0.25$, IID, $K=97$ — too many clients

Key observations from six-panel comparison:

- **Failed runs** show persistently high drift (> 0.8), flat IPR (~ 0.03), and train loss that plateaus. Weight norms never reach grokking-associated magnitudes.

- **Delayed grok** follows the same trajectory as the healthy grok (IPR rises, drift drops, weights plateau) but stretched in time. Final values are identical.
- **Mean client drift** (panel d) separates all four cases most cleanly: $K=97$ failure > 0.8 ; healthy grok < 0.3 .

3 Drift Predicts Grokking Delay, Controlling for α (Figure 2)

3.1 The confound

- Naive analysis: $\rho = 0.70$, $p = 1.6 \times 10^{-29}$ across all 237 runs.
- **But:** 42 of 46 failed runs are $\alpha=0.2$, which fails in the centralized setting too.
- Drift correlates with α because both are controlled by the same experimental knobs.
- **Honest question:** does drift predict T_{grok} *within a fixed* α ?

3.2 Within- α correlation (panel b)

- Within- α Spearman rank correlations between drift and T_{grok} :

α	n	Spearman ρ	p -value	Significance
0.25	47	0.79	$< 10^{-3}$	***
0.30	51	0.70	$< 10^{-3}$	***
0.35	42	0.52	$< 10^{-3}$	***
0.50	51	0.34	< 0.05	*

- Strong near the boundary ($\rho = 0.79$ at $\alpha=0.25$), weakens in the interior ($\rho = 0.34$ at $\alpha=0.5$).
- Expected if drift modulates grokking primarily near criticality.

3.3 Causal chain: $K \rightarrow \text{drift} \rightarrow T_{\text{grok}}$ (panel c)

- At $\alpha=0.3$ (IID, Exp 2): increasing K from 2 to 97 monotonically increases both drift and T_{grok} .
- Heterogeneity is controlled (IID), so the only variable is K .
- $\rho = 1.0$ between drift and T_{grok} across K values.

4 Temporal Ordering: IPR Rise Precedes Drift Drop (Figure 3)

4.1 Question

- Do clients align first, then Fourier features emerge? (drift drops $\rightarrow$ IPR rises)
- Or do Fourier features crystallize first, then clients converge? (IPR rises $\rightarrow$ drift drops)

Figure 6: Mechanistic Comparison of Representative Runs

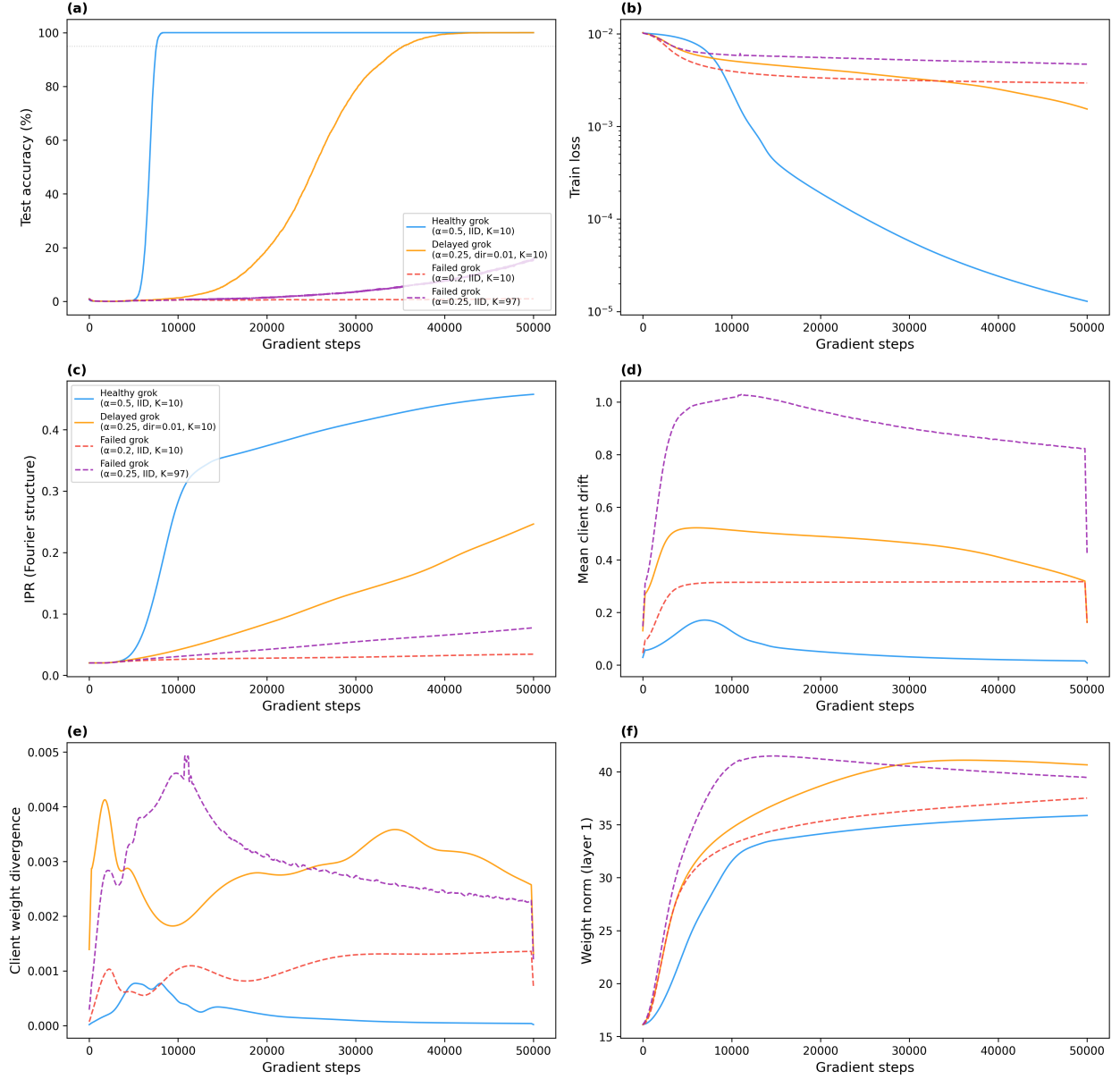


Figure 1: **Representative run comparison.** (a) Test accuracy, (b) train loss (log), (c) IPR, (d) mean client drift, (e) client weight divergence, (f) layer-1 weight norm.

Figure 7: Drift Predicts Grokking Delay (Controlling for α)

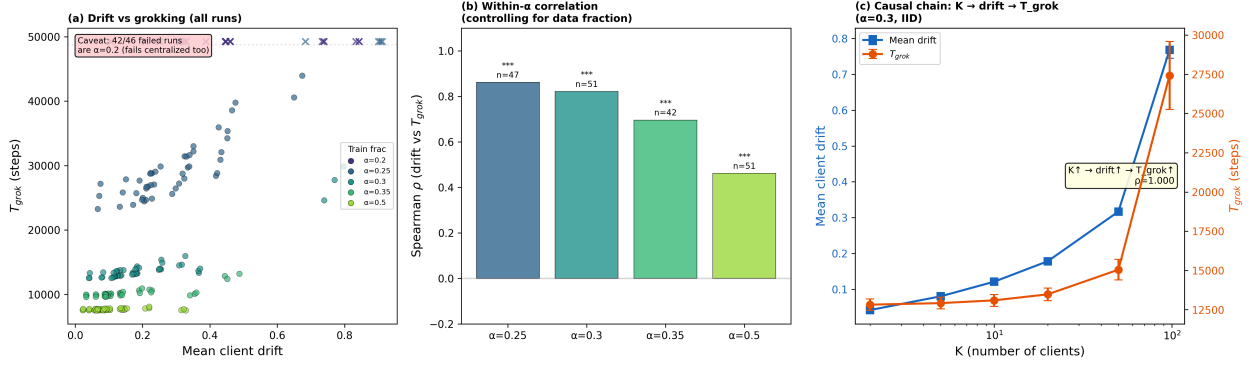


Figure 2: **Drift predicts grokking delay, controlling for α .** (a) T_{grok} vs. drift (all runs; red box: 42/46 failures are $\alpha=0.2$). (b) Within- α Spearman ρ . (c) $K \rightarrow \text{drift} \rightarrow T_{\text{grok}}$ at $\alpha=0.3$.

4.2 Method

- For each of 72 grokking runs (Exp 3a, $K=10$), detect:
 - $T_{\text{IPR onset}}$: first round IPR exceeds $2.5\times$ its early-training baseline
 - $T_{\text{drift drop}}$: first round (post-peak) smoothed drift falls below 50% of its peak

4.3 Results

- IPR onset precedes drift drop in **72 of 72** runs (zero exceptions).
- Median lag: $T_{\text{drift drop}} - T_{\text{IPR onset}} = 10,775$ steps.
- Wilcoxon signed-rank test: $p = 1.8 \times 10^{-13}$.

4.4 Robustness

- Tested 12 combinations: IPR threshold $\in \{2\times, 2.5\times, 3\times, 4\times\} \times \text{drift fraction} \in \{30\%, 50\%, 70\%\}$.
- **All 12 settings:** IPR rises first in 100% of runs (Figure 3d).

4.5 Caveats

- **Temporal precedence $\neq$ causation.** Both events could be driven by a third variable (e.g., weight norm growth reaching a critical value).
- **Threshold asymmetry.** IPR onset = crossing a low absolute value (~ 0.05); drift drop = falling below a fraction of peak. The *ordering* is robust; the *lag magnitude* depends on threshold choice.

Figure 8: Temporal Ordering — IPR Rise Precedes Drift Drop

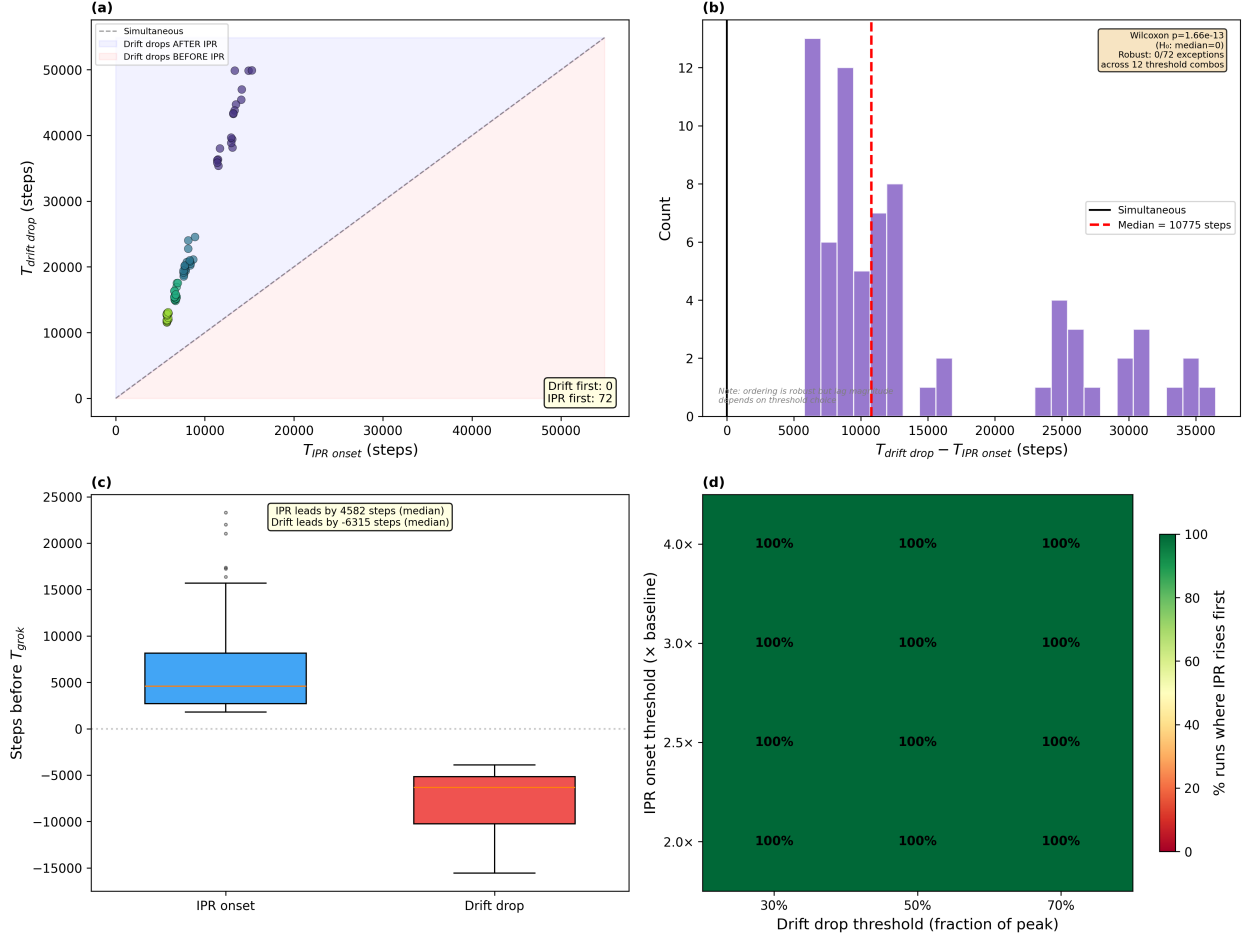


Figure 3: **Temporal ordering.** (a) $T_{\text{IPR onset}}$ vs. $T_{\text{drift drop}}$: all points above diagonal. (b) Lag histogram (median = 10,775 steps). (c) Box plots: both precede T_{grok} ; IPR leads more. (d) Robustness heatmap: 100% IPR-first across all threshold combos.

5 Scaling Collapse (Figure 4)

5.1 Setup

- Rescale time as $\tau = (t - T_{\text{grok}})/(T_{\text{grok}} - T_{50})$, where T_{50} = first step test accuracy reaches 50%.
- Apply to 117 grokking runs from Exps 2–3.

5.2 Accuracy collapse (panel b) — expected

- Test accuracy curves collapse onto a sigmoid (fitted $k \approx 2.48$).
- **Partly tautological:** centering on the transition and scaling by the width will collapse any sigmoid-like curve.
- Confirms transitions are sigmoidal, but not evidence of deeper universality on its own.

5.3 IPR and drift collapse (panels c, d) — non-trivial

- IPR **also collapses** under the same rescaling, despite not being constrained by the procedure:
 - Begins rising at $\tau \approx -3$ (3 transition widths before grokking)
 - Plateaus at $\tau \approx 0$
- Normalized drift follows a universal inverted-U:
 - Peaks at $\tau \approx -2$
 - Declines through the transition
 - Stabilizes at $\sim 50\%$ of peak after grokking ($\tau > 0$)
- **Implication:** feature crystallization, client convergence, and generalization evolve on a single intrinsic timescale that varies across configurations but maintains fixed internal structure.

6 Two-Timescale Separation (Figure 5)

6.1 Definitions

- T_{memo} : first step with train accuracy $\geq 99\%$ (memorization complete)
- T_{grok} : first step with test accuracy $\geq 95\%$ permanently (generalization)
- Generalization delay: $\Delta T = T_{\text{grok}} - T_{\text{memo}}$

Figure 9: Scaling Collapse — 117 Grokking Runs

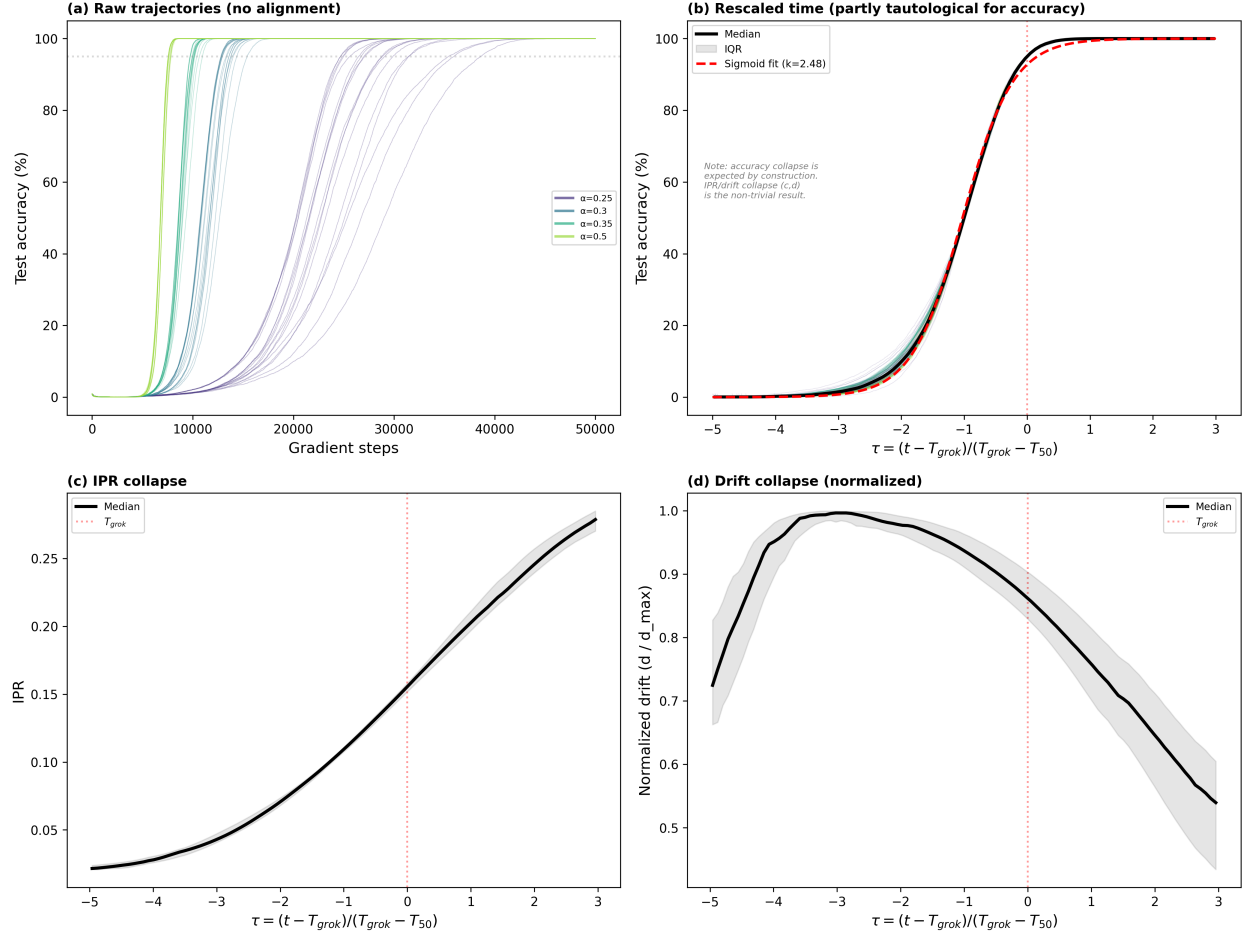


Figure 4: **Scaling collapse.** (a) Raw trajectories (117 runs). (b) Rescaled accuracy (expected for sigmoids). (c) IPR collapse (non-trivial): features emerge at $\tau \approx -3$. (d) Drift collapse (non-trivial): peaks at $\tau \approx -2$. Black = median; shading = IQR.

6.2 Within- α variability (panel a)

- CV of T_{memo} vs. T_{grok} within each α , across all K and heterogeneity:

α	CV(T_{memo})	CV(T_{grok})	Ratio
0.25	0.042	0.159	$4\times$
0.30	0.033	0.068	$2\times$
0.35	0.022	0.026	$1\times$
0.50	0.005	0.005	$1\times$

- Near the boundary ($\alpha=0.25$): T_{grok} is $4\times$ more variable than T_{memo} .
- In the interior ($\alpha \geq 0.35$): both equally stable.

6.3 T_{memo} vs. K (panel b)

- T_{memo} is nearly flat from $K=2$ to $K=50$:
 - $\alpha=0.3$: within $\pm 1\%$
 - $\alpha=0.5$: within $\pm 0.3\%$
 - Only at $K=97$: modest $+17\%$ increase ($\alpha=0.3$)
- T_{grok} diverges:
 - $\alpha=0.3$: 12,822 ($K=2$) $\rightarrow$ 27,422 ($K=97$) — $2.1\times$ slowdown
- **Conclusion:** FL does not impair memorization. It specifically disrupts the transition from memorization to generalization.

6.4 Drift prolongs generalization delay (panel c)

- $\Delta T = T_{\text{grok}} - T_{\text{memo}}$ vs. drift, with within- α Spearman:
 - $\alpha=0.25$: $\rho \approx 0.73$
 - Weakens in the interior, paralleling Section 3.

7 Summary

1. Drift predicts grokking delay within each α .

- Within- α Spearman ρ : 0.34 ($\alpha=0.5$) to 0.79 ($\alpha=0.25$)
- Not a proxy for α ; carries independent predictive information
- *Caveat*: 42/46 failures are $\alpha=0.2$ (fails centralized)

2. IPR rise precedes drift drop (temporal ordering).

- 72/72 runs, 12/12 threshold combinations, zero exceptions
- Consistent with: Fourier features crystallize in global model first, then clients converge

Figure 10: FL Disrupts Generalization, Not Memorization

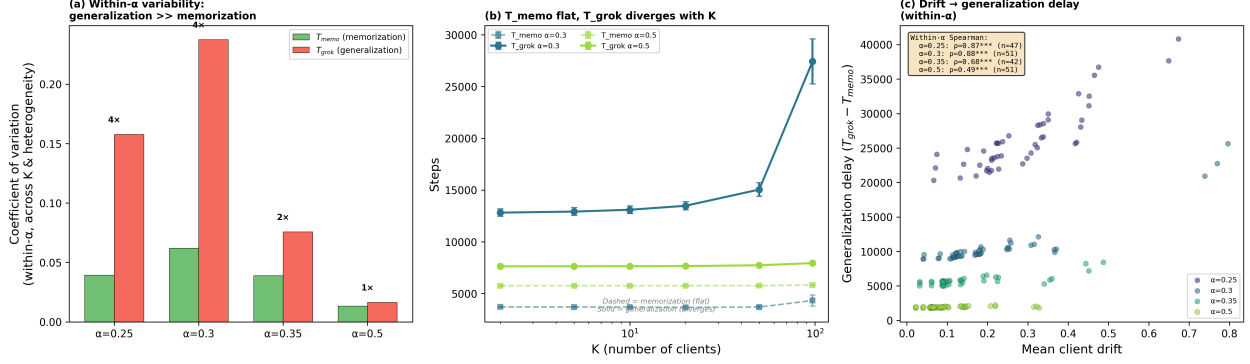


Figure 5: **FL disrupts generalization, not memorization.** (a) Within- α CV: T_{grok} up to 4 $\times$ more variable than T_{memo} . (b) T_{memo} (dashed, flat) vs. T_{grok} (solid, diverging) vs. K . (c) Generalization delay vs. drift with within- α correlations.

- *Caveat*: temporal ordering $\neq$ causation; lag magnitude depends on threshold choice

3. Approximate scaling collapse.

- Under $\tau = (t - T_{\text{grok}})/(T_{\text{grok}} - T_{50})$: test accuracy, IPR, and drift collapse onto shared curves
- Accuracy collapse is partly tautological (any sigmoid collapses)
- IPR and drift collapses are non-trivial $\rightarrow$ single intrinsic timescale governs the transition

4. FL disrupts generalization, not memorization.

- T_{memo} invariant to K (CV $\approx$ 0.005–0.04)
- T_{grok} varies by up to 2.1 $\times$
- Drift specifically prolongs the memorization-to-generalization gap

7.1 Limitations

- **No model checkpoints.** Cannot compute Fourier spectra of W_1 at specific timepoints to test whether clients converge to the same Fourier modes. Requires re-running with checkpoint saving.
- **No per-client trajectories.** Individual client IPR, loss, and weights are aggregated before logging. Cannot distinguish synchronized vs. leader-follower transitions.
- **Limited FL-specific failures.** Only 4/46 failures are FL-caused (not $\alpha=0.2$). Exps 4–5 will provide more.
- **Single architecture and task.** All results are for 2-layer MLP on modular addition ($p=97$). Generalization to other settings is open.